

YOUR FIRST CYBERSECURITY HOMELAB

Build Observe Attack and Understand it

Presented by Justin Wan

liljuicetin.org • homelabvillage.org

About Me

- Research Assistant II at UAH CCRE (2+ years)
- Healthcare IT experience at Broward Health (1.3 years)
- SIEM/SOC experience with CybrHawk (6 months)
- PBSC Ethical Hackers Club alumni

The goal today is to make cybersecurity feel reachable.



Before We Touch the Lab

Use only your assigned lab and workshop-provided credentials.

Do not scan, intercept, or access other attendees, the venue network, or outside systems. Workshop VMs and network activity may be monitored for teaching, troubleshooting, and safety.

Do not use real passwords or personal information. Use throwaway workshop credentials only.

If you reach anything outside your assigned lab, stop and tell me.

By continuing, you agree to stay inside the workshop scope.



Meet Your Lab Environment

BLUE VM (GUI)

Ubuntu Desktop
10.77.X.2
Docker and Pi-hole
tcpdump for visibility

RED VM (CLI)

Attacker and traffic
generator
10.77.X.3
Nmap for scanning
dig and curl for traffic



To Begin: JuiceBeryl Wi-Fi > lab.liljuicetin.org

Recon Before: What Is Already Exposed?

Look at the Blue VM from the attacker point of view before we build anything

```
ssh attacker  
nmap -sV 10.77.X.2  
(Don't know your IP? Run `ip a`)
```

- Which TCP ports are open?
- What services does Nmap identify?
- Keep a mental note of these.
- We will scan again after Pi-hole.



Build: Pi-hole

A real DNS service using Docker Compose

```
cd ~/labs/pihole  
docker compose up -d  
docker ps
```

CLI challenge

```
nano compose.yml  
  
Ctrl+O, Enter = save  
Ctrl+X = exit
```

Beginner path: open compose.yml graphically.
Compose is a consistent configuration for your service.
Useful software can also expose new network services.



Recon After: What Changed?

Same scan. Different attack surface.

```
ssh attacker  
nmap -sV 10.77.X.2
```

BEFORE

Existing services
SSH / RDP / etc.

AFTER PI-HOLE

New DNS / web
service exposure

New service adds a new attack surface.



DNS: Traffic Tells a Story

Red VM generates a DNS request. Blue VM records it.

```
dig @10.77.X.2 example.com
```

- Open the Pi-hole Query Log.
 - Look for the request from 10.77.X.3.
- We generated traffic and then observed evidence.

Attackers generate traffic. Defenders observe evidence.



Packet Capture: HTTP Is Not Private

If the service is not encrypted, application data can be sniffed/read

```
sudo tcpdump -i lo -s 0 -A 'tcp dst port 80'
```

Demo password

[Go crazy]

Readable payload example

```
{"password":"fakepassword","totp":null}
```

Now try it again with HTTPS and port 443



Summary

We just ran a tiny cyber range

BUILD

Pi-hole

OBSERVE

DNS and packets

ATTACK

Nmap and plaintext
traffic

ANALYZE

What changed?

The goal is not memorizing exact commands. It is to understand how self-hosting new services affects the networking side.



Where This Can Take You

The same homelab mindset scales into real security practice

My Home Lab

40+ Docker services
3 Proxmox clusters (1 of which is hosting this workshop)
Project San FranSokyo (Self-hosted SOC-LLM)
SIEM

NEXT PROJECTS

*VPN for secure remote access
*Nginx Proxy Manager
IDS and network monitoring
Active Directory lab
Vulnerable apps and CTFs (such as metasploitable + OWASP Juice Shop)

Today was networking and visibility. Next you can explore logs, identity, web apps, and detection.



Q&A



Thank you!



Justin Wan

Clinical Systems Analyst-Associate | Security+,
Epic Radiant Certified | Member of Palm Beach St...



Workshop slides & resources:

homelabvillage.org/resources

